STAT2004 – Tutorial Week 10 Solutions

Exercises 1 and 2 are covered in the tutorial. Exercise 3 is optional. Exercises 4–6 are homework.*

Exercise 1 (Likelihood ratio test for exponential data)

(a) The claim that "90% of their LED light bulbs last over 24 months before failure" is equivalent to the probability statement $P(X \leq 24) = 0.1$. Using the expression for the cumulative distribution function,

$$0.1 = P(X \leq 24) = F_\lambda(24) = 1 - \mathrm{e}^{-24\lambda} \quad \Longleftrightarrow \quad \lambda = \frac{\ln(10/9)}{24} \approx 0.0044\,.$$

The manufacturer's claim is therefore equivalent to the null hypothesis $H_0 : \lambda = 0.0044$. The one-sided alternative is $H_1 : \lambda > 0.0044$ (a higher rate of failure than claimed).

(b) The likelihood ratio test statistic for testing $\lambda = \lambda_0$ against any $\lambda = \lambda_1 > \lambda_0$ is

$$\frac{L(\lambda_0 \mid X)}{L(\lambda_1 \mid X)} = \frac{\lambda_0}{\lambda_1}\,\mathrm{e}^{(\lambda_1 - \lambda_0)X}\,.$$

Since $\lambda_1 > \lambda_0$, this ratio is *small* when X is small. Thus, the LRT is equivalent to the test with rejection region $\{X \leq d\}$ for some critical value d. This makes intuitive sense: the shorter the observed lifetime X of the light bulb, the more evidence against the null claim that the failure rate is only 0.0044.

(c) To control the Type I error at 5% under $H_0 : \lambda = 0.0044$, choose d such that

$$0.05 = P_{\lambda=0.0044}(X \leq d) = 1 - \mathrm{e}^{-0.0044d} \quad \Longleftrightarrow \quad d \approx 11.68\,.$$

The decision rule is therefore $\{X \leq 11.68\}$ (in months).

(d) Yes. By the Neyman–Pearson lemma, the likelihood ratio test is the most powerful test for testing the simple null $\lambda = \lambda_0$ against any simple alternative $\lambda = \lambda_1 > \lambda_0$. Because the resulting decision rule $\{X \leq 11.68\}$ does not depend on the particular value of λ_1, it is in fact uniformly most powerful for the composite alternative $H_1 : \lambda > \lambda_0$.

(e) The observed outcome is 12 months and 12 days, which is roughly $x_{\text{obs}} = 12.4$ months. This does *not* lie in the rejection region $\{X \leq 11.68\}$, so we retain the null hypothesis $H_0 : \lambda = 0.0044$ at the 5% level. That is, based on this single observation, we accept the manufacturer's claim.

(f) The p-value for this one-sided test is

$$P_{H_0}(X \leq x_{\text{obs}}) = P_{\lambda=0.0044}(X \leq 12.4) = 1 - \mathrm{e}^{-0.0044\times 12.4} \approx 0.053 = 5.3\%\,.$$

If the true failure rate were 0.0044, then observing a lifetime of 12.4 months, or even shorter, would occur by chance about 5.3% of the time. This is not especially rare, so there is only marginal evidence against the manufacturer's claim — consistent with our decision in part (e) to retain H_0.

Exercise 2 (Testing the precision of an apparatus)

(a) $H_0 : \sigma^2 = \sigma_0^2$ versus $H_1 : \sigma^2 > \sigma_0^2$.

(b) Since $X_1, \ldots, X_n$ are iid $N(9.807, \sigma^2)$, the likelihood function is

$$L(\sigma^2 \mid \boldsymbol{X}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - 9.807)^2\right).$$

(c) Consider the likelihood ratio for testing $\sigma^2 = \sigma_0^2$ against $\sigma^2 = \sigma_1^2$ for some $\sigma_1^2 > \sigma_0^2$:

$$\frac{L(\sigma_0^2 \mid \boldsymbol{X})}{L(\sigma_1^2 \mid \boldsymbol{X})} = \left(\frac{\sigma_1^2}{\sigma_0^2}\right)^{n/2} \exp\left(-\frac{1}{2}\left[\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}\right]\sum_{i=1}^n (X_i - 9.807)^2\right).$$

Because $\sigma_1^2 > \sigma_0^2$, this ratio is small when the sum of squares $\sum_{i=1}^n (X_i - 9.807)^2$ is large. Thus, the LRT reduces to the test with rejection region

$$\left\{\sum_{i=1}^n (X_i - 9.807)^2 \geq c\,, \quad \text{for some critical value } c\right\}.$$

(d) Under $H_0 : \sigma^2 = \sigma_0^2$, standard theory for normal samples (with *known* mean 9.807) gives

$$\frac{1}{\sigma_0^2}\sum_{i=1}^n (X_i - 9.807)^2 \sim \chi_n^2\,.$$

(e) Writing $\chi_n^2(0.95)$ for the upper 5%-tile of the χ_n^2 distribution,

$$0.05 = P\left(\chi_n^2 \geq \chi_n^2(0.95)\right) = P_{H_0}\left(\sum_{i=1}^n (X_i - 9.807)^2 \geq \sigma_0^2\,\chi_n^2(0.95)\right).$$

So, to control the Type I error rate at 5%, we should choose $c = \sigma_0^2\,\chi_n^2(0.95)$.

(f) When μ is also unknown, the analogous result for the sample sum of squares about $\bar{X}$ (which loses one degree of freedom for estimating μ) is

$$\frac{1}{\sigma_0^2}\sum_{i=1}^n \left(X_i - \bar{X}\right)^2 \sim \chi_{n-1}^2 \quad \text{under } H_0 : \sigma^2 = \sigma_0^2\,.$$

(g) Similarly to part (e), to control the Type I error rate at 5% we should choose $d = \sigma_0^2\,\chi_{n-1}^2(0.95)$, where $\chi_{n-1}^2(0.95)$ is the upper 5%-tile of the χ_{n-1}^2 distribution.

(h) Here $\sigma_0^2 = 0.02^2$ and $n = 18$, so we need the upper 5%-tile of the χ_{17}^2 distribution:

```
qchisq(0.95, 17)
> 27.58711
```

and so the decision rule for this scenario is

$$\left\{ \sum_{i=1}^{18} \left(X_i - \bar{X}\right)^2 \geq 0.02^2 \times 27.587 \approx 0.105^2 \right\} .$$

In terms of the sample variance this rule is $\left\{ \frac{1}{17} \sum_{i=1}^{18} (X_i - \bar{X})^2 \geq 0.0252^2 \right\}$, which is equivalent to rejecting whenever the sample standard deviation $S \geq 0.0252$ (approximately, $S \geq 0.025$).

(i) If both tests are carried out at the 5% level, the chance that a perfectly good machine *passes* each individual test is 95%. The first test depends only on the sample variance and the second test depends only on the sample mean, and for normal data the sample mean and sample variance are independent. So the chance a perfectly good machine passes *both* tests is $0.95 \times 0.95 = 0.9025$. Thus, the chance that a perfectly good machine *fails* the overall (sequential) test is $1 - 0.9025 = 0.0975 = 9.75\%$ — almost double the nominal 5%.

(j) We want to choose a common level α for the two individual tests so that the overall chance of a perfectly good machine passing both is 95%, i.e. solve $(1-\alpha)^2 = 0.95$. This gives

$$\alpha = 1 - \sqrt{0.95} \approx 0.0253 = 2.53\% .$$

(Note that the simple Bonferroni correction of $5\%/2 = 2.5\%$ is a quick, slightly conservative, way to achieve approximately the same thing.)

Exercise 3* (Sum of Poisson random variables)

(a) A full inductive proof of the binomial theorem is given below for completeness; you may skip straight to parts (b)–(d) if you wish to use the result without proving it.

Base case: The cases $y = 0$ and $y = 1$ are trivial to check directly. (The case $y = 2$, namely $(\lambda_1 + \lambda_2)^2 = \lambda_1^2 + 2\lambda_1\lambda_2 + \lambda_2^2$, also coincides with the given formula.)

Inductive step: Suppose the formula holds for some integer y, i.e.

$$(\lambda_1 + \lambda_2)^y = \lambda_1^y + \binom{y}{1}\lambda_1^{y-1}\lambda_2 + \binom{y}{2}\lambda_1^{y-2}\lambda_2^2 + \cdots + \binom{y}{y-1}\lambda_1\lambda_2^{y-1} + \lambda_2^y . \qquad (*)$$

Now consider $y+1$. Note that

$$(\lambda_1 + \lambda_2)^{y+1} = \lambda_1(\lambda_1 + \lambda_2)^y + (\lambda_1 + \lambda_2)^y\lambda_2 .$$

Substituting $(*)$ into each term and collecting terms of the same order in λ_1, λ_2 gives

$$(\lambda_1 + \lambda_2)^{y+1} = \lambda_1^{y+1} + \sum_{j=1}^{y} \left[\binom{y}{j} + \binom{y}{j-1} \right] \lambda_1^{y+1-j}\lambda_2^j + \lambda_2^{y+1} .$$

By Pascal's rule,

$$\binom{y}{j} + \binom{y}{j-1} = \frac{y!}{(y-j)!j!} + \frac{y!}{(y-j+1)!(j-1)!} = \frac{y!}{(y-j+1)!j!}\big[(y-j+1)+j\big] = \frac{(y+1)!}{(y+1-j)!j!} = \binom{y+1}{j}$$

Hence

$$(\lambda_1 + \lambda_2)^{y+1} = \sum_{j=0}^{y+1} \binom{y+1}{j} \lambda_1^{y+1-j}\lambda_2^j ,$$

which is the same formula with y replaced by $y+1$, completing the induction.

(b) To obtain the event $\{X_1 + X_2 = y\}$, the first observation X_1 can take any value x_1 from 0 to y, but then necessarily $X_2 = y - x_1$. Because X_1 and X_2 are independent, summing over the (disjoint) ways of achieving $X_1 + X_2 = y$ gives

$$P(X_1 + X_2 = y) = \sum_{x_1=0}^{y} \Big[P(X_1 = x_1)\, P(X_2 = y - x_1) \Big], \quad y = 0, 1, 2, \ldots .$$

(c) Starting from the convolution formula in part (b),

$$\begin{aligned} P(X_1 + X_2 = y) &= \sum_{x_1=0}^{y} \Big[P(X_1 = x_1)\, P(X_2 = y - x_1) \Big] \\ &= \sum_{x_1=0}^{y} \mathrm{e}^{-\lambda_1} \frac{\lambda_1^{x_1}}{x_1!}\, \mathrm{e}^{-\lambda_2} \frac{\lambda_2^{y-x_1}}{(y-x_1)!} \\ &= \frac{\mathrm{e}^{-(\lambda_1+\lambda_2)}}{y!} \sum_{x_1=0}^{y} \frac{y!}{x_1!(y-x_1)!}\, \lambda_1^{x_1} \lambda_2^{y-x_1} \\ &= \frac{\mathrm{e}^{-(\lambda_1+\lambda_2)}}{y!} (\lambda_1 + \lambda_2)^y , \end{aligned}$$

using the binomial theorem from part (a). This is exactly the pmf of a Poisson$(\lambda_1 + \lambda_2)$ distribution evaluated at y. Hence $Y = X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

(d) Applying part (c) repeatedly: $X_1 + X_2 \sim \text{Poisson}(\lambda + \lambda) = \text{Poisson}(2\lambda)$; then $(X_1 + X_2) + X_3 \sim \text{Poisson}(2\lambda + \lambda) = \text{Poisson}(3\lambda)$; and so on. By induction, the total count $Y = X_1 + X_2 + \cdots + X_n$ has a Poisson$(n\lambda)$ distribution.

Exercise 4 (Hypothesis tests for the Poisson rate)

(a) $H_0 : \lambda = 7.3$ versus the one-sided alternative $H_1 : \lambda > 7.3$.

(b) The likelihood function is

$$L(\lambda \mid \boldsymbol{X}) = \prod_{i=1}^{15} \frac{\lambda^{X_i}}{X_i!} \mathrm{e}^{-\lambda} \propto \mathrm{e}^{-15\lambda}\, \lambda^{X_1 + \cdots + X_{15}} ,$$

so the log-likelihood is

$$\ell(\lambda \mid \boldsymbol{X}) = -15\lambda + (X_1 + \cdots + X_{15}) \log \lambda + \text{constant} .$$

This is smooth and concave in λ, so setting the derivative to zero,

$$0 = \frac{d\ell}{d\lambda} = -15 + \frac{X_1 + \cdots + X_{15}}{\lambda} ,$$

gives the maximiser $\hat{\lambda}_{\text{MLE}} = (X_1 + \cdots + X_{15})/15 = \bar{X}$, which coincides with the method-of-moments estimator.

(c) The rule $\{\bar{X} \geq c\}$ is equivalent to $\{X_1 + \cdots + X_{15} \geq 15c\}$. Under H_0, and using the result of Exercise 3(d), the total count $X_1 + \cdots + X_{15} \sim \text{Poisson}(15 \times 7.3) = \text{Poisson}(109.5)$. Using software,

$$P(\text{Poisson}(109.5) \geq 127) = 0.055\,, \qquad P(\text{Poisson}(109.5) \geq 128) = 0.045\,.$$

So the decision rule

$$\{X_1 + \cdots + X_{15} \geq 128\} \quad \Longleftrightarrow \quad \{\bar{X} \geq 8.53\}$$

has Type I error just below 5% (the largest achievable level not exceeding 5%, since the test statistic is discrete).

(d) If the true call rate is $\lambda = 9.0$, then $X_1 + \cdots + X_{15} \sim \text{Poisson}(15 \times 9.0) = \text{Poisson}(135)$, so the power of the test from part (c) is

$$P_{\lambda=9.0}(X_1 + \cdots + X_{15} \geq 128) = P(\text{Poisson}(135) \geq 128) \approx 0.738\,.$$

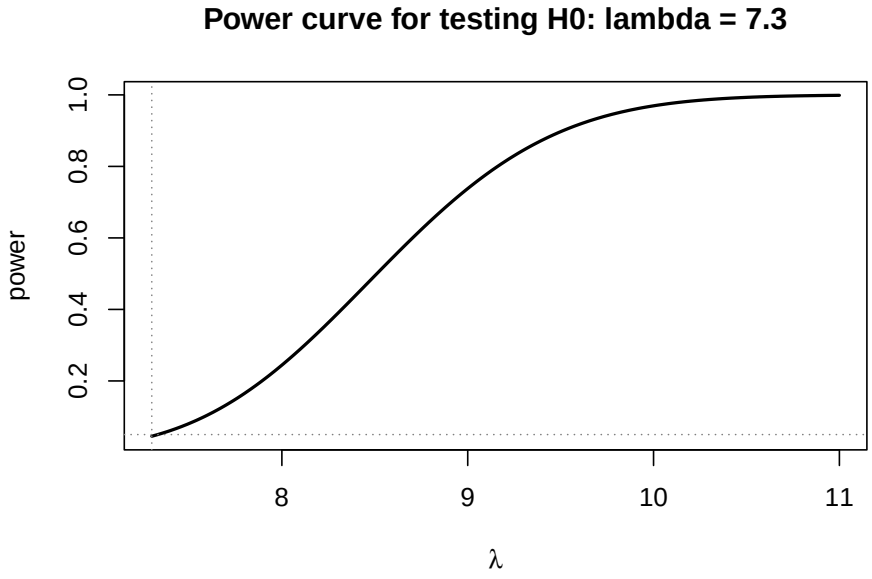
In words: if the true call rate is really 9.0 calls per hour, there is a 73.8% chance that this test correctly rejects the (too low) claimed rate of 7.3 calls per hour.

(e) Repeating the power calculation across a grid of λ values from 7.3 to 11.0 in R:

```
# create a grid of values of lambda from 7.3 to 11.0 in steps of 0.01
lambda = (730:1100)/100

# power of the test "reject H0 if total >= 128" evaluated at each lambda
power = 1 - ppois(127, 15*lambda)

# plot the power curve against lambda:
plot(lambda, power, type="l")
```



(f) The observed total is 132 (so $\bar{x} = 132/15 = 8.8$). Since $132 \geq 128$ (equivalently $\bar{x} = 8.8 \geq 8.53$), we **reject** $H_0 : \lambda = 7.3$ in favour of $H_1 : \lambda > 7.3$: the data support the employee's suspicion that the true call rate is higher than she is being paid for.

(g) The p-value is the probability, under H_0, of observing a test statistic at least as extreme as the one observed:

$$P_{H_0}\big(X_1 + \cdots + X_{15} \geq 132\big) = P\big(\text{Poisson}(109.5) \geq 132\big) \approx 0.020\,.$$

This supplements the decision in part (f): if the call rate really were only 7.3 calls per hour, a total this large (or larger) would occur only about 2% of the time by chance, which is reasonably strong evidence against the null claim.

Exercise 5 (One-sided confidence intervals for the Poisson rate)

Recall that in Exercise 4(f)–(g) we rejected $H_0 : \lambda = 7.3$ in favour of $H_1 : \lambda > 7.3$ at the 5% level, based on the observed total of 132 calls.

(a) If instead $\lambda_0 = 7.5$, then under H_0 the total count $X_1 + \cdots + X_{15} \sim \text{Poisson}(15 \times 7.5) = \text{Poisson}(112.5)$. The p-value for testing $H_0 : \lambda = 7.5$ versus $H_1 : \lambda > 7.5$ is

$$P_{H_0}\big(X_1 + \cdots + X_{15} \geq 132\big) = P(\text{Poisson}(112.5) \geq 132) \approx 0.039\,.$$

Since $0.039 < 0.05$, we would still **reject** $H_0 : \lambda = 7.5$ at the 5% level.

(b) If instead $\lambda_0 = 7.6$, then under H_0, $X_1 + \cdots + X_{15} \sim \text{Poisson}(15 \times 7.6) = \text{Poisson}(114)$, and the p-value is

$$P_{H_0}\big(X_1 + \cdots + X_{15} \geq 132\big) = P(\text{Poisson}(114) \geq 132) \approx 0.053\,.$$

Since $0.053 > 0.05$, we would now **retain** $H_0 : \lambda = 7.6$ at the 5% level.

(c) Repeating these calculations for λ_0 near the boundary between parts (a) and (b):

$$P(\text{Poisson}(15 \times 7.579) \geq 132) \approx 0.04998\,, \qquad P(\text{Poisson}(15 \times 7.580) \geq 132) \approx 0.05012\,.$$

So the largest value of λ_0 that we would still reject at the 5% level is $\lambda_0 = 7.579$ (to 3 decimal places).

Hence any λ_0 in the range $(7.580, \infty)$ would be *retained* in the test of $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda > \lambda_0$ at the 5% level, giving the right one-sided 95% confidence interval $(7.580, \infty)$ — consistent with "we can be 95% confident that the true call rate is at least 7.58 calls per hour".

(d) Repeating the same search but now at the 10% level, the largest λ_0 that we would still reject at the 10% level is $\lambda_0 \approx 7.834$ (to 3 decimal places). So a 90% right one-sided confidence interval for λ, based on the observed data, is

$$(7.834, \infty)\,.$$

We can be 90% confident that the true call rate is at least 7.834 calls per hour. (Note this interval is narrower, i.e. the lower bound is larger, than the 95% interval from part (c) — a lower confidence level gives a less conservative, tighter bound.)

Exercise 6 (Power curves, sample size and relative effect size)

(a) The likelihood function of μ given $\boldsymbol{X} = (X_1, \ldots, X_n)^\top \sim N(\mu, 1)$ iid is

$$L(\mu \mid \boldsymbol{X}) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \exp\left(-\frac{1}{2}\sum_{i=1}^n (X_i - \mu)^2\right).$$

The likelihood ratio statistic for testing $H_0 : \mu = 0$ versus $H_1 : \mu = \mu_1 > 0$ is

$$\begin{aligned}\frac{L(0 \mid \boldsymbol{X})}{L(\mu_1 \mid \boldsymbol{X})} &= \exp\left(-\frac{1}{2}\left[\sum_{i=1}^n X_i^2 - \sum_{i=1}^n (X_i - \mu_1)^2\right]\right) \\ &= \exp\left(-\mu_1 \sum_{i=1}^n X_i + \frac{1}{2} n\mu_1^2\right).\end{aligned}$$

Since $\mu_1 > 0$, this ratio is small when $\sum_{i=1}^n X_i$ is large. Thus, the LRT is equivalent to the test with rejection region

$$\left\{\sum_{i=1}^n X_i \geq d \text{ for some critical value } d\right\} \iff \{\bar{X} \geq c \text{ for some critical value } c\}.$$

(b) Under H_0, $\bar{X} \sim N(0, 1/n)$, i.e. $\sqrt{n}\bar{X} \sim N(0, 1)$. For sample size $n = 5$:

i. We require

$$0.01 = P(N(0,1) \geq 2.326) = P_{H_0}(\sqrt{5}\bar{X} \geq 2.326) = P_{H_0}(\bar{X} \geq 2.326/\sqrt{5}) = P_{H_0}(\bar{X} \geq 1.04),$$

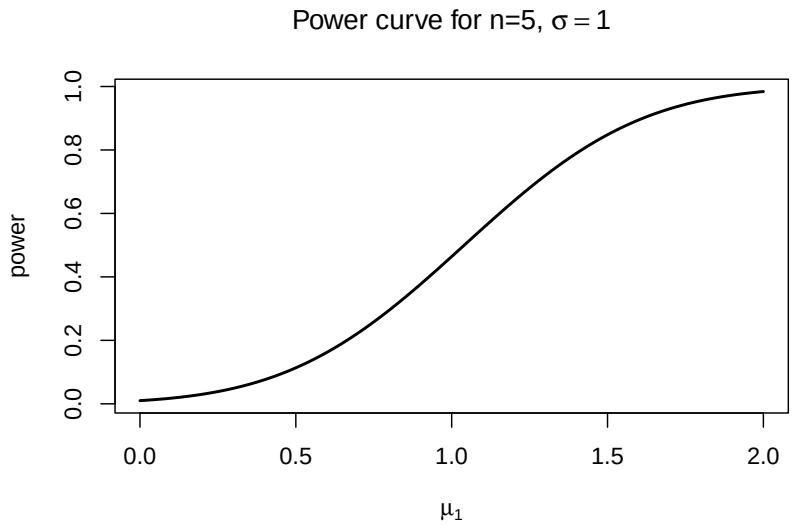
so the cutoff value giving level $\alpha = 0.01$ is $c = 1.04$.

ii. Under the alternative, $\bar{X} \sim N(\mu_1, 1/n)$, i.e. $\sqrt{n}(\bar{X} - \mu_1) \sim N(0, 1)$. The power at alternative value μ_1 is

$$\text{power}(\mu_1) = P_{\mu_1}(\bar{X} \geq 1.04) = P\big(N(0,1) \geq \sqrt{5}(1.04 - \mu_1)\big).$$

Plotting this against $\mu_1 \in [0, 2]$ in R:

```
n = 5
c = 2.326/sqrt(n)
mu1 = (0:200)/100
power5 = 1 - pnorm(sqrt(n)*(c-mu1))
plot(mu1, power5, type="l")
```

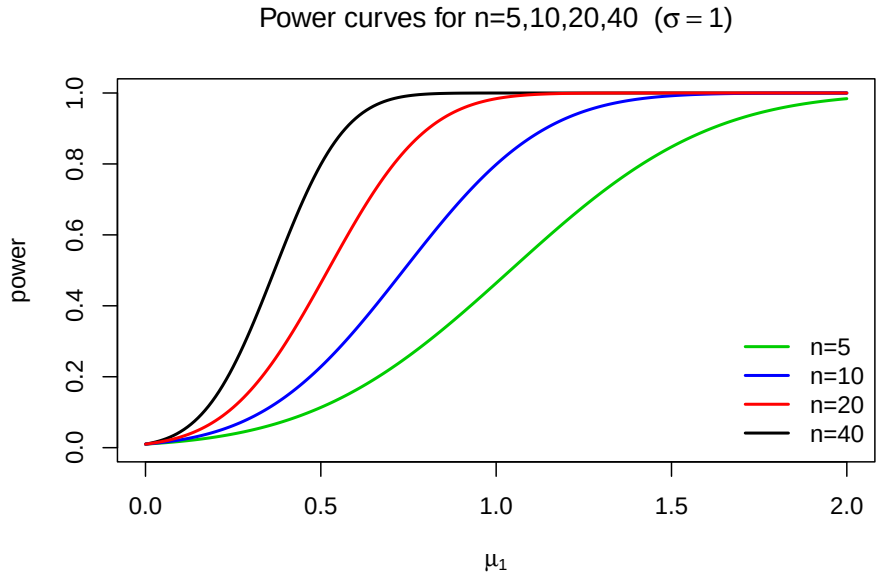


(c) In general, the 1% cutoff for sample size n is $c_n = 2.326/\sqrt{n}$, and the power function is

$$\text{power}(\mu_1) = P\big(N(0,1) \geq \sqrt{n}(c_n - \mu_1)\big)\,.$$

Computing and overlaying the power curves for $n = 5, 10, 20, 40$:

```
plot(mu1, power40, type="l", ylab = "power")
lines(mu1, power20, col="red")
lines(mu1, power10, col="blue")
lines(mu1, power5, col="green")
legend("bottomright", c("n=40","n=20","n=10","n=5"),
       fill = c("black","red","blue","green"))
```

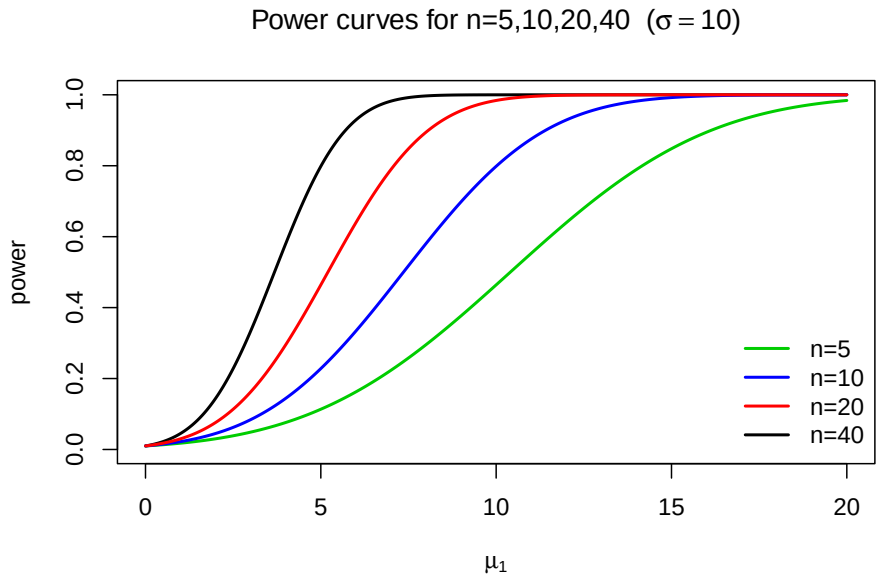


Larger sample sizes give higher power (a steeper curve) for detecting the same alternative μ_1.

(d) Now suppose $X_1, \ldots, X_n$ are iid $N(\mu, 10^2)$. Repeating parts (a)–(c) with $\sigma = 10$: the LRT again reduces to $\{\bar{X} \geq c\}$; under H_0, $\bar{X} \sim N(0, 100/n)$, so the 1% cutoff is now $c_n = 10 \times 2.326/\sqrt{n}$, and the power function is

$$\text{power}(\mu_1) = P\left(N(0,1) \geq \frac{\sqrt{n}(c_n - \mu_1)}{10}\right).$$

Plotting the power curves for $n = 5, 10, 20, 40$ over $0 \leq \mu_1 \leq 20$:



(e) The graphs in parts (c) and (d) look identical in shape, except that the μ_1-axis in part (d) has been stretched (rescaled) by a factor of 10 — exactly matching the tenfold increase in σ. This shows that the power of this test does not depend on the *absolute* effect size μ_1 alone, but on the **relative effect size** (signal-to-noise ratio) $\delta = \mu_1/\sigma$, together with the sample size n.

(f) Here $\sigma = 1.1$ mmol/L and the hypothesised improvement is $\mu_1 = 0.6$ mmol/L, giving a relative effect size

$$\delta = \frac{0.6}{1.1} \approx 0.545.$$

Reading off the power curves from part (c) (which are indexed by δ for $\sigma = 1$), we need the sample size at which the power curve reaches 0.80 at $\mu_1 \approx 0.545$. This occurs at approximately $n = 40$ (more precise computation gives power ≈ 0.87 at $n = 40$ and power ≈ 0.82 at $n = 35$, so n in the mid-to-high 30s is already enough, and $n = 40$ comfortably exceeds 80% power). So we would recommend recruiting around $n = 40$ patients per group to detect this effect with at least 80% power at the $\alpha = 0.01$ level.